

Rithish S

AI Engineer Intern | Autonomous AI Agents (MCP), LLMs/RAG & Scalable Systems

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EDUCATION

Bannari Amman Institute of Technology
Bachelor of Engineering in Computer Science and Engineering
CGPA: 8.28/10

Sathyamangalam, Tamil Nadu
Aug. 2024 – May 2028

TECHNICAL SKILLS

AI, Agents & RAG: Autonomous AI Agents, Model Context Protocol (MCP), Tool Calling, RAG Pipelines, Vector Search (Pinecone), Local LLMs (Qwen2.5, Ollama), Embeddings, Evals & Prompt Engineering
Languages & Core CS: Python, C++, Java, JavaScript (ES6+), TypeScript, SQL, OOP, DBMS, OS, Computer Networks
Backend & Workflows: FastAPI, Node.js, Fastify, Express.js, REST APIs, Redis (Distributed Queues & Locks), BullMQ, Async Pipelines
Cloud, DevOps & Databases: PostgreSQL, MySQL, Redis, AWS (Cloud Practitioner), Oracle Cloud (OCI Certified), Docker, Kubernetes, Linux (POSIX)
Observability & Tools: Git, GitHub Actions, Prometheus, Grafana Telemetry, k6 Performance Testing, AST Static Analysis, Prisma

AI & ENGINEERING PROJECTS

PatentIQ | Autonomous AI Agent & RAG Engine | *Python, Node.js, Fastify, LLMs (Qwen2.5), MCP, Pinecone, Prisma* [GitHub](#)

- Architected an **autonomous AI agent** workflow using the **Model Context Protocol (MCP)** and dynamic **tool calling**, dynamically orchestrating vector retrieval tools, patent claim parsers, and external research APIs.
- Engineered a low-latency **RAG** pipeline utilizing local **LLMs (Qwen2.5 via Ollama)** and Pinecone vector embeddings to automate unstructured data analysis, semantic claim overlap, and quantitative novelty risk scoring.
- Built a modular Fastify and PostgreSQL backend with Clean Architecture, dependency injection, and schema validation, providing structured JSON outputs for multi-step agentic execution loops and high API reliability.

DB Backup CLI (DBVault) | Distributed Disaster Recovery & Queues | *TypeScript, Node.js, Redis, BullMQ, AWS S3, Docker* [GitHub](#)

- Architected an automated multi-engine backup/restore CLI published on npm and Docker Hub with **120 automated tests**, supporting PostgreSQL, MySQL, MongoDB, and SQLite with automated scheduling and checksum validation.
- Designed an asynchronous distributed job execution system using **Redis and BullMQ**, enforcing worker concurrency, rate-limits, execution guards, and Redis-based distributed locking to guarantee zero data loss.
- Implemented authenticated AES-256-GCM encryption with local key rotation, stream-based compression, and automated cloud offloading via modular **AWS S3** object storage adapters.

EXPERIENCE

Web Developer Feb. 2025 – Dec. 2025
Students Special Group *Student Organization*

- Collaborated with cross-functional student teams to engineer a full-stack **Project Management Portal** with React, Node.js, and MySQL, streamlining day-to-day sprint task tracking and workflow progress.
- Assisted with data analysis and reporting by designing REST APIs and automated data aggregation models for a **Faculty Reward Points Dashboard**, delivering real-time metric visualization and performance reporting.
- Participated in team technical reviews, contributing ideas on continuous software improvement while mentoring **50+ students** in Data Structures, backend system design, and collaborative Git workflows.

OPEN SOURCE & CODE QUALITY

ESLint Core | *JavaScript, AST Static Analysis, IEEE 754 Floating-Point Precision* [PR #21218](#)

- Contributed a merged fix to ESLint's **no-loss-of-precision** rule, resolving subtle IEEE 754 binary64 floating-point underflow edge cases to distinguish non-zero decimals from zero literals during AST static analysis.
- Engineered comprehensive regression test suites covering subnormal numbers, signed zero, and non-decimal literals, successfully passing **43 automated CI verification checks** in a world-standard open-source project.

ACHIEVEMENTS & CERTIFICATIONS

- 1200+ Algorithmic Problems Solved** across LeetCode, CodeChef, and GeeksforGeeks, focusing on Dynamic Programming, Graph algorithms, Trees, and analytical problem-solving.
- IEEE DevSpark Hackathon Winner** – Secured **1st place among 120+ teams** by developing **Plantera**, an AI-powered environmental monitoring platform with WebGL spatial visualization.
- BIT Hackathon Winner** – Secured **1st place among 200+ teams** by building a full-stack Leave Management workflow system under 24-hour sprint constraints.
- Official Certifications:** IIT Madras Python: Elite + Topper (Top 5%, 95% score) (NPTEL26CS84S454501242); Oracle Cloud Infrastructure 2025 Certified Foundations Associate (1030565840CI25FNDCA); AWS Cloud Practitioner Essentials; Cisco Linux.